

Activity 1

0.05

Aim: To prepare 100 mL of 0.1 M standard solution of oxalic acid.

Theory: The molecular formula of oxalic acid is $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$. Its molar mass is 126 g/mol.

1000 mL of 1M solution of oxalic acid contains 126 g of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$

Hence 100 mL of 0.1M solution of oxalic acid will contain

$$= \frac{100 \text{ mL} \times 0.1 \text{ M} \times 126 \text{ g}}{1000 \text{ mL} \times 1 \text{ M}} = 1.26 \text{ g of oxalic acid.}$$

Apparatus: 100 mL volumetric flask, beaker, glass rod etc.

Chemicals: Oxalic acid, distilled water.

Procedure:

1. Weigh accurately 1.26 g of oxalic acid. Transfer it to a beaker and add little distilled water. Stir it well with the help of glass rod.
2. Transfer the solution from beaker to volumetric flask, using conical funnel and glass rod properly (Take care to prevent loss of even a single drop of solution).
3. Wash the beaker twice / thrice with distilled water and transfer the washing to the volumetric flask.
4. Dilute the solution by adding distilled water up to the mark on the volumetric flask to make the volume 100 mL. (See that the volume marking appears tangential to the meniscus of the final solution)

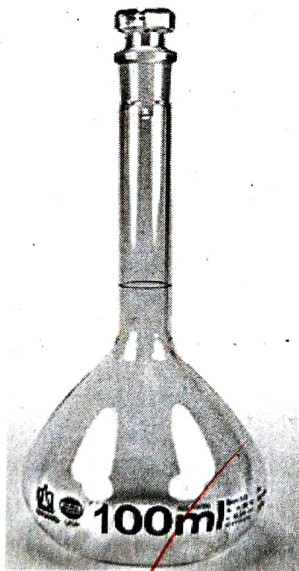


Fig. Activity 1.1 Volumetric flask.

Conclusion: Thus, 100 mL 0.05 M standard solution oxalic acid is prepared.

Remark and sign of teacher:

V. Good

Korant

28/11/24

MCQ

Select the correct answer for the following multiple choice type of questions.

1. A solution whose concentration is accurately known is called-----
a. dilute solution b. concentrated solution
☒ c. standard solution d. colloidal solution
2. The molar mass of oxalic acid is
a. 126 u ☒ b. 126 g / mol
c. 12.6 u d. 12.6 g / mol
3. To prepare standard 1M solution of oxalic acid.
a. 0.126 g of oxalic acid is dissolved in 1 L of solution.
b. 1.26 g of oxalic acid is dissolved in 1 L of solution.
c. 12.6 g of oxalic acid is dissolved in 1 L of solution.
☒ d. 126 g of oxalic acid is dissolved in 1 L of solution.
4. To prepare 100 mL of standard 0.1M oxalic acid solution.
a. 0.63 g of oxalic acid is dissolved in 100 ml of distilled water.
☒ b. 1.26 g of oxalic acid is dissolved in 100 ml of distilled water.
c. 12.6 g of oxalic acid is dissolved in 1000 ml of distilled water.
d. 126 g of oxalic acid is dissolved in 100 ml of distilled water

Short answer questions

1. Calculate the amount of oxalic acid required to prepare 1000 mL 0.1 M standard solution of oxalic acid

Ans.

126 g of oxalic acid in 1000 ml = 1M

$\therefore$ 12.6 g of oxalic acid in 100 ml = 0.1M

2. Calculate the molar mass of oxalic acid. (C = 12, O = 16, H = 1)

Ans.

Molar mass of Oxalic acid

= $H_2C_2O_4 \cdot 2H_2O$

= $2(H) + 2(C) + 4(O) + 4(H) + 2(O)$

= $2(1) + 2(12) + 4(16) + 4(1) + 2(16)$

= 126 g/mol

Remark and sign of teacher:

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